

AnonDHT in the Boss Fight Regime (Addendum): Supporting Evidence Tables and Pattern Map

February 28, 2026 (archive v125)

Abstract

This companion note collects supporting evidence tables for the Boss Fight dial sheet (Paper B). To keep the main dial sheet referee-readable and compact, we move (i) the compact pattern map to classic AnonDHT designs, (ii) contact/hop amplification tables under representative Kademlia parameters, and (iii) auxiliary dummy-width requirement tables into this addendum. All content is pure $\text{\LaTeX}$ and requires no external artifacts.

Scope. Addendum to the Boss Fight dial sheet.[2]

Imports. We import the Boss Fight accountant and MaxL-to-knob compilation viewpoint from Paper B.[2] We also import the trace tiering interface (for interpreting what a “contact” witness means).[11]

Exports. Supporting tables that a referee or auditor can quote while keeping the main paper short.

IPFS lookup mini-spec slice (iterative). The spec defines an *iterative* lookup: the initiator maintains a candidate list of peers close to the target identifier, sends queries to up to α unqueried candidates, and expands the list using returned peer sets of size k . Termination is reached once the β closest reachable peers to the target have been successfully queried (or the candidate set is exhausted) [7]. In our Boss Fight framing, each queried peer (and each timeout/abort) is a potential destination-indexed witness.

Operational note (deployments and attacks; pointers). We keep this dial sheet focused on MaxL-to-knob compilation rather than a deployment survey. For deployment context, reader-privacy upgrade proposals, and live-network attack/measurement evidence in Kademlia-derived systems (IPFS/libp2p and related overlays), see the profiling paper [4] and the trace-interface note [11]; this dial sheet uses only the transcript shape and representative parameter scales (Table 2, included below). In what follows we use only the iterative-lookup transcript shape and representative parameter scales (Table 2), which determine how many transcript contacts a single lookup can expose under retries/timeouts.

Table 4 illustrates this amplification and its impact on dummy-width requirements under a representative per-lookup budget.

$$p_{\text{obs}} \approx 1 - (1 - \rho)^s. \tag{1}$$

Design / paper	Where it sits in the dial sheet (Boss Fight view)
NISAN [9]	Uses a structured overlay to distribute network information for relay selection; aims for scalability but inherits DHT transcript leakage channels and is stressed by selective failure and repeated queries.
Torsk [8]	Directory/selection replacement using a DHT; incorporates proxy/buddy ideas and secure-state assumptions. The attack analysis in [13] highlights how “one proxy” mechanisms can collapse under repetition and range estimation.
Attacks on NISAN/Torsk [13]	Canonical demonstration that passive range estimation and active selective failures reduce effective anonymity of DHT-based relay selection; motivates explicit budgeting for retries/aborts and higher-order witness constraints (Paper A).
Octopus [14]	“Secure+anonymous lookup” via attacker identification, query/path splitting, and dummy queries; improved anonymity but with nontrivial bandwidth/latency overhead.
IRIS [3]	Tunable privacy-preserving search for authenticated Chord overlays; illustrates that privacy goals must be sized against a concrete repetition window and authentication constraints.
Query obfuscation [6]	Increases per-contact ambiguity by replacing one queried identifier with a small set of candidates (e.g., double hashing); trades extra contacts for reduced per-contact leakage and must be sized under the same accountant.

Table 1: A minimal map from classic AnonDHT designs to our dial sheet. The goal is not to re-survey these systems, but to show which knobs they instantiate and why Boss Fight budgeting makes their trade-offs explicit.

References

- [1] Anonymous. Boss Fight Budgets for Metadata-Hiding Overlays: MaxL/PML, Repetition Windows, and Necessary Witness Conditions. Boss Fight series Paper A, 2026. (This archive.)
- [2] Anonymous. AnonDHT in the Boss Fight Regime: Compiling a MaxL Budget into Lookup Knobs. Boss Fight series Paper B, 2026. (This archive.)
- [3] A. Aktypi and K. Rasmussen. Iris: Dynamic Privacy Preserving Search in Authenticated Chord Peer-to-Peer Networks. In *Network and Distributed System Security Symposium (NDSS)*, 2025. DOI:10.14722/ndss.2025.240392. (Open copy: <https://www.ndss-symposium.org/wp-content/uploads/2025-392-paper.pdf>.)
- [4] Anonymous. Anonymous DHT Profiling and the Equalization Contract. Anonymity series Paper B, 2026. (This archive.)
- [5] M. Cortes-Goicoechea, C. Kiraly, D. Ryajov, J. L. Muñoz-Tapia, and L. Bautista-Gomez. Scalability limitations of Kademlia DHTs when enabling Data Availability Sampling in Ethereum. 2024. ACM DOI:10.1145/3688225.3688236. Open preprint: arXiv:2402.09993. <https://arxiv.org/abs/2402.09993>.

Parameter	Representative scale (libp2p/IPFS-style Kademlia)
Bucket size k	$k = 20$ (canonical Kademlia default; libp2p/IPFS)
Parallelism α	$\alpha = 10$ (common libp2p configuration; implementation-dependent)
Termination threshold β	$\beta = 3$ (lookup stops after β closest reachable peers queried)
Distinct queried peers per lookup s	Typical $s \leq 18$ (99% in one public measurement); rare tail ≈ 100

Table 2: Representative public Kademlia parameters and observed contact counts used for the contact-amplification tables in this addendum. These are not normative; they are a typical scale used to illustrate why the Boss Fight regime bites for iterative lookups.

Operation / setting	Distinct queried peers m (typical scale)
Iterative Kademlia lookup (libp2p/IPFS impl; experiment)	$m \leq 18$ for 99% of lookups under high concurrency; rare tail ≈ 100 [5]
Provider-record advertisement / PUT	Touches at least the k closest peers (IPFS default $k=20$); in ProbeLab’s RFM17, successful contacts have 5th/50th/90th percentiles 14/18/20 [10], and the walk/backtracking can touch more peers [7, 12]

Table 3: Back-of-the-envelope contact counts for Kademlia-style operations. Here “hops” in the measurement source correspond to distinct queried peers. These contact counts drive observation amplification ($p_{\text{obs}} = 1 - (1 - \rho)^s$; see Boss Fight A/B[1, 2]) and make the Boss Fight regime unavoidable unless the transcript is equalized or cryptographically shielded.

- [6] E. Daniel, G. Michel, and F. Tschorsch. Improving Kademlia lookup privacy through query obfuscation. In *ACM/SIGAPP Symposium on Applied Computing (SAC)*, 2025. DOI:10.1145/3672608.3707703. Open PDF: <https://guillaume.michel.id/docs/kademlia-query-obfuscation.pdf>.
- [7] IPFS Standards. IPFS Kademlia DHT. 20 November 2025. <https://specs.ipfs.tech/routing/kad-dht/>.
- [8] J. McLachlan, A. Tran, N. Hopper, and Y. Kim. Scalable onion routing with Torsk. In *ACM CCS*, 2009. DOI:10.1145/1653662.1653733.
- [9] A. Panchenko, S. Richter, and A. Rache. NISAN: Network information service for anonymization networks. In *ACM CCS*, 2009. DOI:10.1145/1653662.1653681. (Open copy: <https://www.freehaven.net/anonbib/cache/ccs09-nisan.pdf>.)
- [10] ProbeLab. RFM 17: Provider Record Liveness (in-the-wild IPFS DHT measurements; 18 Aug 2022). <https://raw.githubusercontent.com/probe-lab/network-measurements/main/results/rfm17-provider-record-liveness.md>. (Accessed February 2026.)

Table 4: Contact amplification in Kademia-style lookups. For corruption fraction ρ and s distinct queried peers, $p_{\text{obs}} = 1 - (1 - \rho)^s$ is the probability at least one queried peer is adversarial. Using the forced-destination-with- w -dummies slice (Boss Fight A/B)[1, 2], the last column reports the required dummy fraction w/K to achieve a per-lookup budget $b = 0.01$ bits when the effective candidate set is $K = 2^{20}$.

ρ	s	p_{obs}	required w/K
0.01	20	0.182	0.963
0.01	40	0.331	0.979
0.01	80	0.552	0.988
0.01	120	0.701	0.990
0.05	20	0.642	0.989
0.05	40	0.871	0.992
0.05	80	0.983	0.993
0.05	120	0.998	0.993
0.10	20	0.878	0.992
0.10	40	0.985	0.993
0.10	80	1.000	0.993
0.10	120	1.000	0.993
0.20	20	0.988	0.993
0.20	40	1.000	0.993
0.20	80	1.000	0.993
0.20	120	1.000	0.993

- [11] Anonymous. Trace interfaces for anonymous DHTs: observation tiers, committee-contact traces, and exposure normal forms. Synthesis series note, 2026. (This archive.)
- [12] D. Trautwein, Y. Wei, Y. Psaras, M. Schubotz, I. Castro, B. Gipp, and G. Tyson. IPFS in the Fast Lane: Accelerating Record Storage with Optimistic Provide. *IEEE INFOCOM*, 2024. DOI:10.1109/INFOCOM52122.2024.10621404. Open PDF: <https://www.fiz-karlsruhe.de/sites/default/files/FIZ/Dokumente/Forschung/Mathematik/INFOCOM24-IPFS.pdf>.
- [13] Q. Wang, P. Mittal, and N. Borisov. In search of an anonymous and secure lookup: attacks on structured peer-to-peer anonymous communication systems. In *ACM CCS*, 2010. DOI:10.1145/1866307.1866343. (Open copy: <https://www.freehaven.net/anonbib/cache/ccs10-lookup.pdf>.)
- [14] Q. Wang and N. Borisov. Octopus: A secure and anonymous DHT lookup. In *IEEE ICDCS*, 2012. DOI:10.1109/ICDCS.2012.78.

Table 5: Committee observation probability under independent node corruption. If each committee of size s contains a corrupted observer with probability $q = 1 - (1 - \rho)^s$ when the global corruption rate is ρ , then even moderate ρ yields large q , making per-identifier visibility high.

ρ	$s = 3$	$s = 5$	$s = 7$	$s = 9$
0.05	0.143	0.226	0.302	0.370
0.10	0.271	0.410	0.522	0.613
0.20	0.488	0.672	0.790	0.866
0.30	0.657	0.832	0.918	0.960
0.40	0.784	0.922	0.972	0.990

Table 6: Committee corruption $\rightarrow$ observation probability $\rightarrow$ dummy-cover pressure. We take $K = 2^{20}$ plausible targets and a Boss Fight window with $Q = 1000$ lookups and guess cap $\delta = 10^{-3}$, giving $b = \log_2(\delta K)/Q \approx 0.01$ bits per lookup (Paper A). We approximate committee observability by $p_{\text{obs}} \approx 1 - (1 - \rho)^s$ (Eq. (1)), then compute the minimum w satisfying the forced-destination-with- w -dummies slice (Boss Fight A/B)[1, 2]. Even modest corruption and committee sizes push w/K close to 1.

ρ	s	$p_{\text{obs}} \approx 1 - (1 - \rho)^s$	min. w	w/K
0.1	5	0.410	1,031,063	0.983
0.1	10	0.651	1,037,496	0.989
0.1	15	0.794	1,039,471	0.991
0.2	5	0.672	1,037,838	0.990
0.2	10	0.893	1,040,468	0.992
0.2	15	0.965	1,041,070	0.993
0.3	5	0.832	1,039,881	0.992
0.3	10	0.972	1,041,123	0.993
0.3	15	0.995	1,041,298	0.993

Table 7: Dummy width required to achieve a per-lookup MaxL budget under the forced-destination+ w -dummies pattern (Paper A) with observation probability $p_{\text{obs}} = 0.9$. Values are the smallest integers w satisfying $\mathcal{L} \leq b$. When w is comparable to K , dummy cover becomes an impractical knob.

K destinations	$b=0.25$	$b=0.5$	$b=1$	$b=2$
2^{16}	54,151	44,880	31,043	15,123
2^{20}	866,426	718,086	496,693	241,979

Table 8: When per-lookup budgets are in the 10^{-3} – 10^{-2} bit range (as forced by repetition), dummy cover becomes essentially the entire keyspace. We use the conservative mixture bound $\mathcal{L} \leq \log_2((1 - p_{\text{obs}}) + p_{\text{obs}}K/(w+1))$ and solve for the smallest integer w such that $\mathcal{L} \leq b$, with $K = 2^{20}$ and $p_{\text{obs}} = 0.9$.

b (bits)	required w
0.001	1,047,768
0.005	1,044,546
0.010	1,040,534
0.050	1,009,038
0.100	971,130